

# Utility Theory and Decision Networks

Lecture 22

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CS 4880

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# Last Time: The Robot Estimated Its State

Dynamic Bayesian networks and filters gave the robot a belief about the current world:

$$bel_t(x) = P(X_t = x \mid z_{1:t}).$$

- ▶ HMMs represented compact discrete hidden state.
- ▶ Kalman filters represented linear-Gaussian continuous state.
- ▶ Particle filters represented flexible beliefs using weighted samples.

## Lecture 21 question

“Where am I, and what is happening around me?”

## Key idea

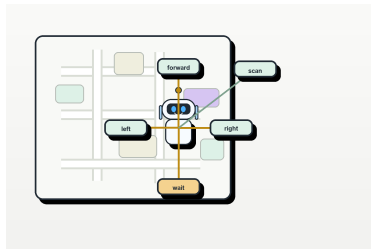
State estimation gives the agent beliefs about the world; decision theory uses those beliefs to choose actions.

# The Robot Is No Longer Following a Fixed Route

The delivery robot no longer receives a precomputed sequence of moves.

At every control cycle it must choose what to do next based on:

- ▶ its current position belief;
- ▶ uncertain obstacles and pedestrian traffic;
- ▶ remaining delivery time;
- ▶ battery state;
- ▶ current task priority.



## Key idea

The planning problem has moved inside the loop: estimate the current situation, choose the next action, observe the result, repeat.

# A Good State Estimate Still Does Not Tell Us What to Do

Suppose the robot believes

$$P(\textit{Blocked}) = 0.35, \quad P(\textit{Clear}) = 0.65.$$

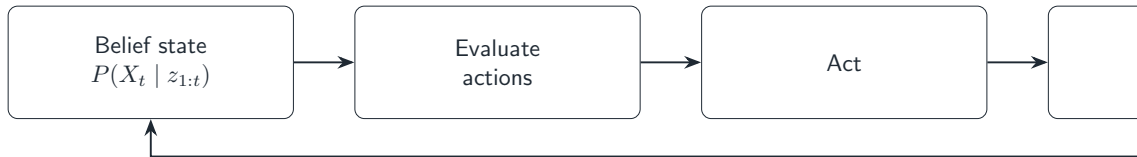
That belief does not answer:

- ▶ Is the shortcut worth attempting?
- ▶ Should the robot take a longer but safer detour?
- ▶ Is it worth waiting for congestion to clear?
- ▶ Is another sensor scan worth the time and energy?

## Key idea

Probability ranks how plausible outcomes are. Decision making also requires preferences over those outcomes.

# The New Control Loop



The robot repeatedly converts a current belief into a next action.

## Key idea

Decision theory turns the robot from a passive estimator into an agent that continually converts beliefs into actions.

# At One Campus Intersection, the Robot Has Several Actions

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## Shortcut

Fastest route. May be blocked by a crowd or maintenance cart.

## Detour

Longer route around the building. Usually clear and predictable.

## Wait

Spend time while congestion may clear.

## Scan

Pause for a better sensor reading before committing.

## Key idea

An action is valuable because of the distribution of outcomes it creates—not because its name sounds desirable.

# Probability Alone Cannot Choose Between Actions

The current belief is

$$P(Clear) = 0.65, \quad P(Blocked) = 0.35.$$

But should the robot take the shortcut?

We still need to know:

- ▶ how valuable a fast successful shortcut is;
- ▶ how costly a blocked attempt is;
- ▶ how much delay the detour causes;
- ▶ how strongly the system prefers safety, time, and energy conservation.

## Key idea

Choosing an action requires both a probabilistic model of consequences and a preference model over those consequences.

# Utility Turns Preferences into a Numerical Model

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A **utility function** assigns a numerical value to an outcome:

$$U(o) : outcome \rightarrow \mathbb{R}.$$

Example preference ordering:

$$U(\text{fast safe delivery}) > U(\text{safe detour}) > U(\text{blocked attempt}).$$

## Key idea

Utility is a compact numerical representation of the agent's preferences over outcomes.

# Only Relative Utility Matters

Utility is not a physical measurement like meters or joules.

If a positive affine transformation is applied,

$$U'(o) = aU(o) + b, \quad a > 0,$$

the same preference ordering is preserved.

Example:

$$U(\text{fast safe}) = 80, \quad U(\text{detour}) = 45, \quad U(\text{blocked}) = -60.$$

## Key idea

Utility values are meaningful through preference comparisons and tradeoffs, not as standalone units.

# Under Uncertainty, Use Expected Utility

For an action  $a$  with possible outcomes  $o$ ,

$$EU(a) = \sum_o P(o \mid a)U(o).$$

A rational expected-utility decision rule chooses

$$a^* = \arg \max_a EU(a).$$

## Probability

How likely is each outcome if I choose this action?

## Utility

How desirable is each outcome?

## Decision

Choose the action with greatest expected utility.

## Key idea

Expected utility weights each possible consequence by both its probability and its desirability.

# Worked Decision: Shortcut or Detour?

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Current belief:

$$P(\textit{Clear}) = 0.65, \quad P(\textit{Blocked}) = 0.35.$$

Utility model:

$$\begin{aligned} U(\textit{Shortcut}, \textit{Clear}) &= 80, \\ U(\textit{Shortcut}, \textit{Blocked}) &= -60, \\ U(\textit{Detour}) &= 45. \end{aligned}$$

## Key idea

The action with the most likely good outcome is not necessarily the action with the highest expected utility.

## Expected Utility of the Shortcut

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$$\begin{aligned} EU(\textit{Shortcut}) &= 0.65(80) + 0.35(-60) \\ &= 52 - 21 = 31. \end{aligned}$$

The shortcut is more likely to be clear than blocked, but the blocked outcome is costly enough to reduce its expected utility substantially.

### Key idea

Expected utility combines the upside and downside of an action in one probability-weighted quantity.

# Expected Utility of the Detour

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For the simplified example, the detour is predictable:

$$EU(Detour) = 45.$$

Therefore

$$45 > 31.$$

The rational one-step choice is the detour.

## Key idea

Rational choice is about expected preference, not simply choosing the most probable outcome.

# Most Likely Outcome $\neq$ Best Action

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## Maximum-probability thinking

"The shortcut is 65% likely to be clear, so take the shortcut."  
This ignores how bad the 35% failure case is.

## Expected-utility thinking

"The shortcut is often faster, but failure is costly enough that the detour has higher expected utility."

## Key idea

Decision theory separates probability from preference so rare but severe outcomes can matter appropriately.

# A Robot Usually Has More Than One Objective

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The robot may simultaneously care about:

- ▶ travel time;
- ▶ energy consumption;
- ▶ collision and pedestrian risk;
- ▶ progress toward completing the delivery;
- ▶ reliability and schedule slack.

These objectives can conflict.

## Key idea

Real decision systems often optimize several competing attributes, so the utility model must make those tradeoffs explicit.

# One Engineering Approximation: Multiattribute Utility

A simple model is a weighted sum:

$$U(o) = w_t u_t(o) + w_e u_e(o) + w_s u_s(o) + w_p u_p(o).$$

The weights encode the relative importance of time, energy, safety, and task progress.

## Important modeling assumption

A weighted sum is convenient, but it assumes the attributes can be combined in this way. Real preference interactions may require a richer utility model.

## Key idea

Utility design is a modeling choice: convenient mathematical structure buys simplicity but may hide real preference interactions.

# Changing the Weights Can Change the Robot's Behavior

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## Time-heavy

Large penalty for delay. The robot accepts more uncertainty to preserve deadline slack.

## Energy-heavy

Battery is scarce. The robot avoids steep ramps, long scans, and unnecessary detours.

## Safety-heavy

The robot prefers conservative routes through uncertain pedestrian traffic.

## Key idea

The same probability model can produce very different behavior under different utility functions.

# Not Every Requirement Should Be “Just Another Weight”

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Weighted utilities make tradeoffs possible by design.

But some requirements may instead be treated as **hard constraints**:

- ▶ exclude actions that violate safety envelopes;
- ▶ enforce legal or operational limits;
- ▶ optimize utility only among feasible actions.

This matters when a tiny probability of a catastrophic outcome should not be casually exchanged for a small improvement in speed.

## Key idea

Utility maximization is only as sensible as the objectives, constraints, and action set supplied to it.

# Decision Networks Add Actions and Preferences to Bayes Nets

A **decision network** or **influence diagram** contains three node types:

## Chance node

Uncertain random variable.  
Example: shortcut blocked?

## Decision node

Choice controlled by the agent.  
Example: shortcut, detour, scan, wait.

## Utility node

Preference over resulting outcomes.  
Example: time + energy + safety + progress.

## Key idea

A decision network combines uncertainty, controllable choices, and utility in one graphical model.

# Robot Decision Network: One Next-Move Choice



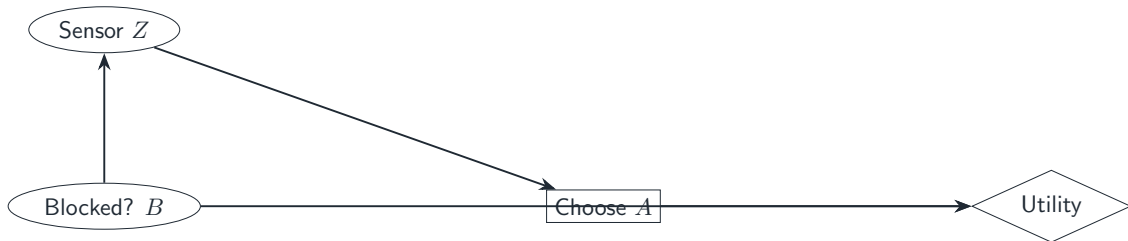
The world state affects the outcome. The chosen action also affects the outcome. Utility evaluates the combination.

## Key idea

Decision networks make the separation between “what the world may do” and “what the agent may choose” explicit.

# Information Arcs Mean “Known Before the Decision”

Suppose a sensor reading  $Z$  is available before the action is chosen.



An arrow into a decision node indicates information available at decision time; it is not a CPT for the decision itself.

## Key idea

The decision can condition on observations the agent has already seen.

# Evaluating a Decision Network: Sum at Chance, Max at Decision

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At a chance variable, average over possible outcomes:

$$EU(a) = \sum_o P(o \mid a)U(o).$$

At the decision variable, choose the maximizing action:

$$a^* = \arg \max_a EU(a).$$

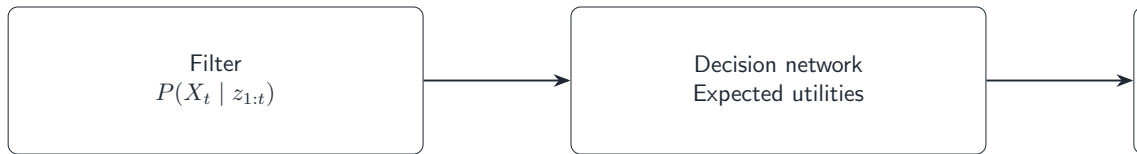
The result is a decision rule or policy for the current decision context.

## Key idea

Probabilistic inference supplies the distribution; decision optimization selects the maximizing action.

# The Belief State from Lecture 21 Plugs Directly Into the Decision

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State estimation and decision making are not competing algorithms. The current belief is an input to the decision.

## Key idea

The agent acts on what it currently believes—not on an inaccessible perfect state of the world.

# One-Step Decision Procedure

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At each control cycle:

1. update the belief state  $b_t$ ;
2. generate feasible actions  $A(b_t)$ ;
3. predict possible outcomes for each action;
4. compute expected utility for each action;
5. execute  $\arg \max_a EU(a)$ .

## Key idea

This is a rational one-step controller: repeatedly choose the action with the best expected immediate outcome under the current belief.

# Sometimes “Get More Information” Is Itself an Action

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## Act now

Choose shortcut or detour using the current belief.

No sensing delay, but uncertainty remains high.

## Scan first

Pause for a better sensor reading, update the belief, then choose.

Costs time and energy, but may improve the eventual action.

## Key idea

Information has instrumental value when it can improve the action the robot chooses afterward.

# Value of Information: Will a Better Measurement Change the Decision?

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Current belief:

$$P(Blocked) = 0.35.$$

Without another measurement, the robot chooses the detour with utility 45.

A short active scan costs 5 utility units and has sensor model

$$P(+ \mid Blocked) = 0.90, \quad P(+ \mid Clear) = 0.10.$$

Is the scan worth its cost?

## Key idea

The value of information depends on how the observation changes downstream decisions, not only on sensor accuracy.

# The Scan Produces Two Possible Belief Updates

First,

$$P(+) = 0.90(0.35) + 0.10(0.65) = 0.38.$$

Then Bayes' rule gives

$$P(\textit{Blocked} \mid +) = \frac{0.90(0.35)}{0.38} \approx 0.829,$$

while

$$P(\textit{Blocked} \mid -) = \frac{0.10(0.35)}{0.62} \approx 0.056.$$

## Key idea

A useful observation partitions the future decision into different belief states that may justify different actions.

# After the Scan, the Best Action Depends on the Observation

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If the scan is positive:

$$P(Blocked \mid +) \approx 0.829,$$

so the shortcut has expected utility approximately  $-36$  and the robot chooses the detour.

If the scan is negative:

$$P(Blocked \mid -) \approx 0.056,$$

so the shortcut has expected utility approximately  $72.1$  and the robot chooses the shortcut.

## Key idea

The scan is valuable because different observations lead to different actions.

## Expected Value of the Scan

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Before subtracting sensing cost,

$$EV(Scan) = 0.38(45) + 0.62(72.1) \approx 61.8.$$

Subtract the cost of scanning:

$$61.8 - 5 = 56.8.$$

Without scanning, the best action has value 45.

Therefore the net value of scanning is

$$56.8 - 45 = +11.8.$$

### Key idea

Information is worth acquiring when its expected improvement in decision quality exceeds its sensing cost.

# Information Can Be Accurate and Still Have Zero Decision Value

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## Observation changes the action

Positive → detour

Negative → shortcut

The information can have positive decision value.

## Same action either way

Positive → detour

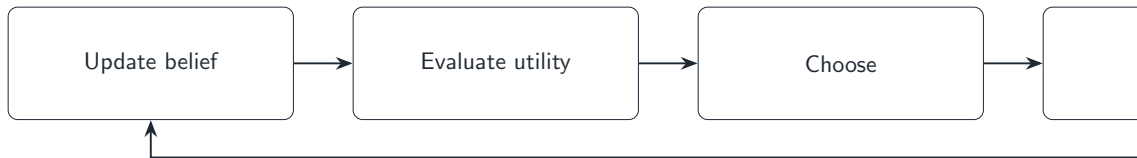
Negative → detour

Even a very accurate observation may have little decision value.

## Key idea

Decision-relevant information is valuable because of its effect on action choice, not because reducing uncertainty is automatically useful.

## Now Put the Decision Back Inside the Robot Loop



This loop may execute dozens or hundreds of times during one delivery.

### Key idea

A decision network can be reevaluated every control cycle using the robot's newest belief state.

# This Gives Us a Myopic Decision Controller

## One-step controller

```
while delivery not complete:
    b_t = update_belief(sensor_data)
    actions = feasible_actions(b_t)
    for a in actions:
        EU[a] = expected_immediate_utility(a,b_t)
    execute(argmax_a EU[a])
```

It is **myopic** because it optimizes what is best now without yet including the value of future states and future actions.

## Key idea

Repeated one-step optimization is a useful reactive strategy, but repeated local optimality does not guarantee globally good behavior.

# Failure Mode: Locally Rational Actions Can Oscillate

Imagine two similar corridors whose estimated congestion changes slightly with each noisy scan.

1. At time  $t$ , left has slightly higher immediate expected utility  $\rightarrow$  turn left.
2. At time  $t + 1$ , a new reading makes right slightly better  $\rightarrow$  turn right.
3. At time  $t + 2$ , noise flips the preference again  $\rightarrow$  turn left.

$$Left \leftrightarrow Right \leftrightarrow Left \leftrightarrow \dots$$

## Key idea

A controller that only asks “what is best this instant?” can thrash when actions have switching costs or delayed effects.

# Failure Mode: The Best Immediate Move Can Create a Bad Future

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## Narrow shortcut

Immediate utility: +30.

Moves quickly toward the destination, but enters a courtyard with one exit and frequent congestion.

## Wider detour

Immediate utility: +20.

Looks worse right now, but preserves several future route choices.

## Key idea

Actions do not merely produce immediate outcomes—they change which states and choices will be available later.

# The Question Has Changed Again

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So far the robot chooses

$$a_t^* = \arg \max_a EU(a \mid b_t).$$

But repeated autonomous control asks something stronger:

**What action is best when I also care about the future states and future decisions it creates?**

## Key idea

Once actions influence future opportunities, rational decision making must reason across sequences of states and actions.

# Next: Markov Decision Processes

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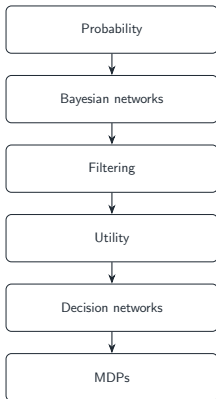
Lecture 23 will formalize the repeated decision problem with:

- ▶ **states**: what situation is the robot in?
- ▶ **actions**: what can the robot do?
- ▶ **transitions**: where might each action take it?
- ▶ **rewards**: what immediate signal does a transition produce?
- ▶ **policy**: what should the robot do in every state?

## Key idea

Decision networks choose under uncertainty; MDPs will formalize repeated decisions whose consequences propagate into the future.

# The Whole Unit Now Forms One Pipeline



## Key idea

The course is progressively adding what an autonomous agent needs: beliefs, time, preferences, actions, and finally long-term strategy.

# Takeaways

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- ▶ Probability describes uncertain outcomes; utility describes preferences over those outcomes.
- ▶ Expected utility combines probability and preference: choose the action with maximum expected utility.
- ▶ Multiattribute utilities make tradeoffs explicit, but weights and modeling assumptions directly shape behavior.
- ▶ Decision networks combine chance variables, decision variables, information, and utility.
- ▶ Value of information asks whether better evidence is worth its sensing cost because it changes downstream decisions.
- ▶ Repeating a decision network every cycle produces a useful myopic controller, but immediate optimality can fail when actions change future states.

## Key idea

The robot can now decide what to do next. Lecture 23 asks how it should decide when “next” changes everything that comes after.